Harsh Meel
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EDUCATION

†Translated

Cornell University, Ithaca, United States — Aug 2024 – Exp. July 2026
Master of Science (Research) in Mechanical Eng.; Conc: Dynamics and Control, Atmospheric Sci. (2.793/4.3)

Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India — 2019 - 2023
Bachelor of Technology in Civil Engineering with Minor in Mechanical Engineering (8.21/10 or 3.420/4.0†)

SKILLS AND RELEVANT COURSES UNDERTAKEN

**Ongoing*

Software	Python, MATLAB, SolidWorks, AutoCAD, Ansys, STAAD, PTV VISSIM, Earth System Models*, Pyomo
Prototyping	Electro-Mechanical Assembly, 3D Printing, Metal and Composite Milling and Fabrication, Adhesives
Selected Courses	Control Systems, Env. Statistics and Learning, Earth System Modeling*, Climate Dynamics, Fundam. of Urban Sci., Renewable Energy Tech., Water Supply and Resources Eng., Manufacturing Process

KEY EXPERIENCES

Regionally Informed Robust Global Climate Engineering, Climate Eng. Lab | MS Thesis [Oct'24- Current]
Principal Advisor: Prof. Douglas MacMartin, MAE, Cornell University; Minor Advisor: Prof. Daniele Visioni, CALS

- Researching uncertainty management in regional precipitation systems due to climate change using Stratospheric Aerosol Injection; Using decision trees, data analytics, dynamics, and feedback control
- Assessing the Indian Monsoons as test case; Aiming for first-author publication to inform public policy

Optimization Tools for Electric Vehicle (EV) Charging Infrastructure| Research Assistant [Nov'23 –April'24]
Guided By: Prof. Ashutosh Mahajan, Industrial Engineering and Operations Research, IIT Bombay, India

- Pointed out opportunities and limitations of using optimization tools in aiding the Indian EV charging infrastructure using cost optimization; Conducted field research and built test case for lab's software

Accumulator (Battery) Design Engineer, IIT Bombay Racing | Formula Student [Reports] [Nov'19-Jun'22]
3-tier cross-functional team of 70+ undergrad students; Winner Formula Bharat'21 Overall and Eng. Design

- Responsible for design, manufacturing, and testing of a 400V, 7.8 kWh LiPo accumulator (battery)
- Designed new parallel configuration of cells, theor. cutting battery heat by 50% and volume by 33%
- Improved battery size estimation using driver mix data and transient simulation of car - increasing reliability and performance; initiated technical alliance with cell recyclers to handle discarded cells
- Trained and managed 3 junior engineers; led engineering recruitment tests for 140+ new aspirants

Wind-Test Setup, ideaForge Technology Ltd., Mumbai, India | R&D intern [May'22 – June'22]
Industry leader in UAV technology in India | Developing drone solutions for Defense, Surveying, and Enterprises

- Designed plan for a modular wind-test setup producing realistic wind profiles to test multiple drones
- Analyzed axial fan flow to reduce total power load; work received a recommendation from the CTO

RELEVANT PROJECTS

Feedback control of simplified Earth's surface temperature | Course Project [Report] [Aug'25-Dec'25]

- Designed a PID controller stabilizing against ramped disturbance(emissions) in presence of time delay

Reliability Analysis of Renewable Grid Transition Using Data-Driven Models [Report] [Nov'24 – Dec'24]
Guided by: Prof. Scott Steinschneider, Biological and Environmental Engineering, Cornell University

- To investigate correlation between price spike events and renewable grid expansion in NY state CLCPA, analyzed load and fuel mix data using regression, decision trees, and ensemble methods

Vehicle Overtake Assist System | Institute Technical Summer Project [Repository] [May'20 – Aug'20]
Led a first-year undergraduate student summer project, Awarded Special Mention (top 7) out of 64+ projects

- Designed a driver assistance system, assisting the vehicle in overtake maneuvers on undivided two-laned roads in poor sight conditions; conducted on-road calibration and virtual simulation testing

Last Updated: April 16th, 2026